

EEG Signal Data Collection

and Random Forest-Based Imputation for Brain-State Analysis

Aditya Krishna — Roll No: 225CS2009

Guide: Prof. Pabitra Mohan Khillar

Department of Computer Science and Engineering, NIT Rourkela

May 7, 2026

- 01 Introduction to EEG Signals
- 02 Brain Frequency Waves
- 03 Electrode Placement (10-20)
- 04 Diseases & EEG Patterns
- 05 Data Collection Hardware
- 06 Signal Processing Pipeline
- 07 Dataset Overview (NIT Rourkela)
- 08 Kaggle BRMH Dataset
- 09 Epilepsy Temple University Dataset
- 10 Data Imputation Techniques
- 11 HTI-MissForest Results
- 12 Future Scope & Conclusion

What is EEG?

Electroencephalography (EEG) records the electrical activity generated by neurons in the brain using scalp electrodes.

- Signals measured in **microvolts (μV)** — very tiny!
- High **temporal resolution** (milliseconds)
- Non-invasive, portable, and affordable
- Captures both **normal and abnormal** brain rhythms

Applications:

- Brain-Computer Interface
- Epilepsy Detection



EEG recording with gold-cup electrodes

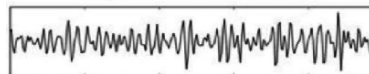
Brain Frequency Waves

Wave	Freq (Hz)	Brain State
Delta	0.5–4	Deep Sleep
Theta	4–8	Meditation
Alpha	8–13	Relaxation
Beta	13–30	Active Thinking
Gamma	30–45	High Cognition

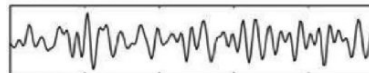
Key Insight

Each brain state produces a **distinct frequency signature**. Monitoring these bands reveals cognitive and clinical conditions.

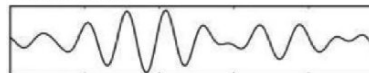
Comparison of EEG Bands



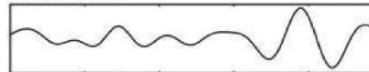
Gamma: 30-100+ Hz



Beta: 12-30 Hz



Alpha: 8-12 Hz



Theta: 4-7 Hz

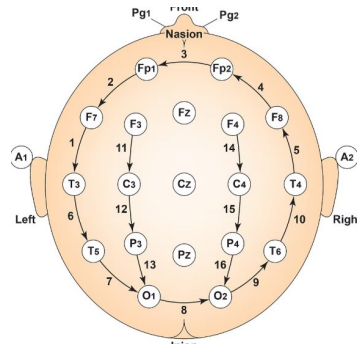


Delta: 0-4 Hz

Comparison of EEG frequency bands

Electrode Placement — International 10-20 System

- Standard used worldwide for EEG recordings
- Electrodes named by **region + hemisphere**:
 - **Fp** = Frontal Pole
 - **F** = Frontal, **C** = Central
 - **P** = Parietal, **O** = Occipital
 - **T** = Temporal
- **Odd numbers** = Left hemisphere
- **Even numbers** = Right hemisphere
- **z** suffix = Midline



International 10-20 electrode positions

This Project Used

Active: Fp1, Fp2, Cz, O1, O2

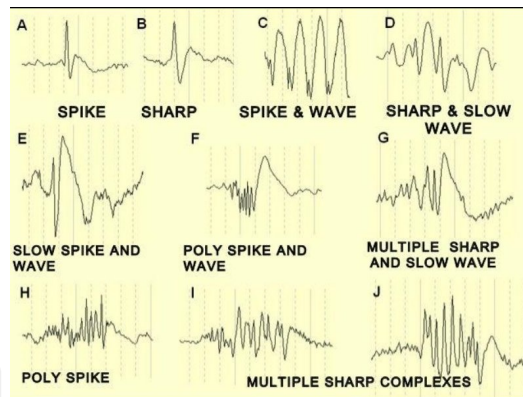
Reference: Left earlobe Ground: Right earlobe

Brain Diseases Detected by EEG

Disease	Key Wave	EEG Observation
Epilepsy	Delta/Spike	Sudden spike bursts
Alzheimer's	Theta	Diffuse slow waves
ADHD	Theta	Excess theta activity
Anxiety	Beta	High-frequency beta
Depression	Alpha	Alpha asymmetry
Sleep Disorder	Delta	Disrupted delta waves
Addiction	All bands	Multi-band disruption

Clinical Relevance

EEG is the **gold standard** for epilepsy diagnosis and plays a growing role in neurological and psychiatric screening.



Epileptic spike-wave patterns

EEG Data Collection Setup

Hardware Components:

- **EWS All-in-one Arduino Mega IoT Grove Kit**
- Gold-cup scalp electrodes (conductive gel)
- INA128 Instrumentation Amplifier (gain $\approx 1000\times$)
- Hardware band-pass filter: 0.5–45 Hz
- USB serial @ 115200 baud to laptop

Data Acquisition:

- Sampling rate: **256 Hz**
- ADC: **10-bit** on analog pin A0
- Python script: timestamps + CSV logging
- Total cost: **Rs. 4,000**

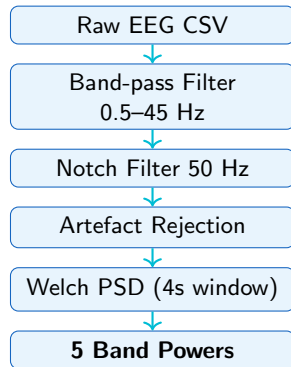


Processing Steps:

- 1 **Band-pass Filter** (0.5–45 Hz, 4th-order Butterworth)
- 2 **Notch Filter** (50 Hz, $Q=30$ — removes mains hum)
- 3 **Artefact Rejection** ($|x| > 150 \mu V \rightarrow$ linear interpolation)
- 4 **Welch PSD** (4-second windows, Hann, 50% overlap)
- 5 **Band Power Integration** (trapezoidal rule per band)

Extracted Features (per second):

- δ Delta, θ Theta, α Alpha, β Beta, γ Gamma
- Units: $\mu V^2/Hz$



Collection Details:

- **30 subjects** recorded at NIT Rourkela
- **117 CSV source files** merged
- **7,787 total records** (1 Hz epochs)
- 4 distinct **brain states**: Learning, Maths, Meditation, Song Listening
- 5 band features per record

Band Power Statistics (μV^2):

Band	Mean	Median	Std
Delta	60.45	0.68	273.78
Theta	96.31	13.47	356.95
Alpha	13.22	4.33	52.36
Beta	11.21	5.96	39.71

Dataset Summary

Total Records: **7,787**

Subjects: **30**

Source Files: **117**

Brain States: **4**

Features: **5 bands**

Sampling: **256 Hz**

High std-dev in Delta & Theta indicates transient burst activity across subjects and tasks.

Dataset Information:

- Source: **Kaggle** — *EEG Psychiatric Disorders Dataset*
- Contains **945 patients** with clinical EEG recordings
- **19-channel** EEG using standard 10-20 placement
- Features: Absolute/Relative band power + Coherence values
- **Total features:** 1000+ columns per record

Disorders Covered:

- Addictive disorder (Alcohol use)
- Anxiety, Depression, PTSD
- Trauma & Stress-related disorders
- Schizophrenia, Mood disorders

BRMH Dataset Stats

945	Subjects
19	Channels
6	Frequency bands
1024	Feature columns
8	Disorder classes

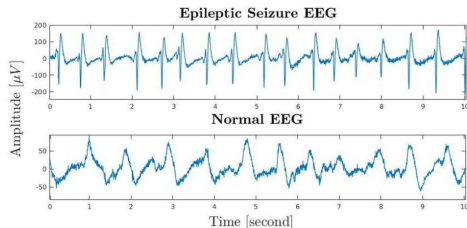
Bands: δ , θ , α , β , highbeta, γ

Dataset Information:

- One of the **largest public EEG datasets** for epilepsy
- Collected at **Temple University Hospital**, Philadelphia, USA
- Over **10,000 EEG recordings** from real patients
- Annotated seizure events with start/end times
- Used for training seizure **detection algorithms**

Why It Matters for This Work:

- Real clinical data has **frequent missing segments**



Epileptic spike-wave patterns from clinical EEG recordings

Seizure EEG vs Normal EEG

Seizure EEG shows **high-amplitude spikes** ($> 150 \mu V$) with erratic discharge patterns, unlike smooth normal rhythms.

Why Data Imputation is Necessary

Causes of Missing Data:

- Electrode lifting during movement
- Motion artefacts (subject moves)
- Serial communication packet drops
- Skin conductance changes
- Hardware/sensor noise bursts

Problems with Naive Fixes:

- **Delete window:** Loses real data, biases results
- **Column mean:** Distorts frequency distribution
- **KNN:** Fails with 5+ correlated band features
- None preserve **inter-band relationships**

Benefits of Proper Imputation:

- Preserves signal **frequency structure**
- Maintains band-power **correlations**
- Enables reliable **ML classification**
- Reduces need for repeated sessions
- Suitable for real-time clinical use

Comparison of Methods

Method	Accuracy	Bands
Mean	Low	No
KNN	Moderate	No
RF-MICE	98.6%	Yes

HTI-MissForest Imputation Pipeline

HTI-MissForest Steps:

1 Feature Engineering

Add: Total Power, α/β Ratio, θ/γ Ratio

2 Temporal Smoothing

2-sample causal rolling mean

3 5% MCAR Masking

Simulate realistic data loss

4 StandardScaler

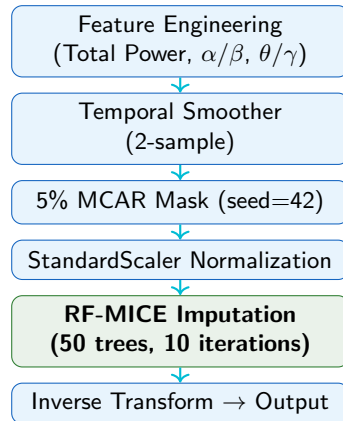
Normalize to zero mean, unit variance

5 RF-MICE Imputation

50 trees, depth 10, up to 10 iterations

6 Inverse Transform

Restore original $\mu V^2/Hz$ scale



Evaluation Protocol:

- **5% MCAR** data removed artificially
- Errors computed **only on masked positions**
- Fixed seed (42) for full reproducibility
- Runs on **Google Colab free tier**

Metric	Value
RMSE	30.23
MAE	5.65
R ² Score	0.9861
Accuracy	98.61%

===== FINAL RESULTS =====					
	Method	RMSE	MAE	R2 Score	Accuracy (%)
0	Random Forest MICE	30.232275	5.649644	0.986133	98.613271

Final output: Random Forest MICE on test set

Significance

An R² of **0.9861** means the model explains **98.6%** of the variance in the missing values — near-perfect recovery.

Hardware Improvements:

- **Multi-channel EEG:** Extend to 16–32 electrode channels
- **Wireless:** Replace USB with Bluetooth/Wi-Fi
- **Edge Deployment:** Run imputation on Raspberry Pi 4
- **Remote Monitoring:** Use onboard SIM7672S cellular modem

Software Improvements:

- **Deep Learning:** LSTM/Transformer for time-series-aware imputation
- **Clinical Validation:** Test on Temple University epilepsy data

Classification Pipeline (Next Step):

- 1 Collect clean, imputed EEG band-power data
- 2 Extract features (ratios, statistics)
- 3 Train classifiers: SVM, Random Forest, CNN
- 4 Predict brain states in real-time
- 5 Detect disorders: Epilepsy, ADHD, Depression

End Goal

A complete **affordable EEG pipeline** — from raw brain signal to clinical diagnosis — accessible in rural Indian health centres.

Conclusion

98.61%

Imputation Accuracy

$R^2 = 0.9861$

Variance Explained

Rs. < 4,000

Full Hardware Cost

- 1 Designed a **low-cost EEG system** (Rs. 4,000) using Arduino Mega IoT Grove Kit.
- 2 Recorded EEG at **256 Hz** from 30 subjects across 4 brain states.
- 3 Developed **HTI-MissForest** with inter-band feature engineering.
- 4 Achieved **98.61% accuracy**, $R^2 = 0.9861$ on 5% MCAR holdout.
- 5 Missing EEG data **reliably recovered** without signal distortion.



Thank You

Questions & Discussion

Aditya Krishna

Roll No: 225CS2009

Guide: Prof. Pabitra Mohan Khilar

Department of Computer Science & Engineering

National Institute of Technology Rourkela